A Value Decomposition for Unforecastable Supply Disruptions

information, routing, inventory — and what simple distributor rules achieve against the RL claim

(deterministic studies on the thesis simulator; no RL; every number audited against raw run data)

Aidan Hansen

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Summary

Across two regimes (the thesis scenario, where the surviving chain has headroom; and a documented saturated extension, where it doesn't) and three echelon levers, the value decomposition is now complete and stress-tested. **(1) Information is the binding constraint in the thesis-severity regime**: a two-line reroute on the *shared upstream machine-state* signal reproduces the perfect-onset oracle bit-for-bit (1.06M → 295k; lost patients 445 → 203); downstream delivery statistics are structurally blind for ∼6 periods (the dead distributor serves from stock). Verification corrected our own earlier claim: with the buffer *correctly located*, true pre-onset foresight is worth ∼54k (∼5%), not zero. **(2) From the distributor's seat — the levers the thesis DRL actually controls — simple rules beat base stock decisively**: a 3-parameter compound (*serve the captive HC while your manufacturer is down* × *stop ordering from the dead factory* × *standing buffer*) removes **49–51% of base stock's episode-profit loss** in the thesis's own world and metric, with better fill and fewer lost patients — roughly *half* of the thesis RL's claimed 89% (an unreplicated number whose baseline configuration we still need from the author). Demand-shaping through the trust loop is confirmed real by a direction control. **(3) Under scarcity, one composition owns the frontier** (buffer-at-the-surviving-chain + reroute + taper: −79%, fill 0.99–1.00; lost patients −68%), and **demand-aware buffer sizing** is the robust knob — sizing B to (demand − surviving deliverable) × duration cuts lost patients to **41 vs. 505** at the frozen-grid optimum. **(4) Two clean negatives sharpen the design rules**: severity-aware redirect caps fail in both natural forms (orders are the supplier's production signal — capping them throttles the ramp), and the buffer's optimal *location* flips to the disrupted chain at severe scarcity (−30% vs. the healthy location). All phases were independently audited from raw CSVs (12/12, 11/12 with one display-rounding correction, 16/16).

Baselines	fairness-neutral (proportional) and fairness-costed (static priority), both carried; thesis-world comparisons use the as-shipped configuration
Seeds	tuning 1–10 (all parameter choices), reporting 11–30 (all reported numbers), paired; RNG-reseed fix gate-guarded
Scoring	system-level, phase-stratified; dual accounting (full vs. excluding manufacturer-queue bookkeeping); thesis-metric (episode PSC profit) for DS-seat claims
Gates	bit-exact baseline reproduction (library AND CLI paths), determinism, conservation, cross-seed variance; two run-batches were discarded after gates/diagnostics caught contamination (CLI defaults) and a bootstrap deadlock — both incidents logged
Audit	every phase re-derived from raw CSVs by an independent agent (no shared analysis code); one display-rounding over-claim caught and corrected

1 Protocol

2 The thesis-severity regime: information, then almost nothing

The surviving manufacturer ramps with incoming orders up to a 400/period cap (∼3.3× its normal load), so rerouting alone restores full service — *if* it happens at onset. The deployable compound (upstream-signal reroute + order taper + priority allocation) reaches 273k against the 1.06M routing-fixed baseline. Verification (A1) rebuilt the oracle as a true superset and corrected our earlier claim:

Table 1: Corrected foresight decomposition (slack regime, urgent0, reporting seeds).

	dp-cost	**increment**
deployable compound (signal reroute + taper)	273,088	—
+ JIT buffer at the HEALTHY DS, zero lead (deployable)	236,762	−36k
+ 10-period pre-onset start (true foresight)	182,222	−55k

The bounded statement of the residual: no rule in the enumerated family {buffer (size×location×timing), taper, throttle, SS-freeze, reroute (trigger×share), allocation (12 rules)} leaves measurable room above the best composition; a learned policy would need a mechanism none of these expresses, and we identify no candidate — but the family bound is the claim, and A1 itself shows why (one mis-located lever had hidden ∼90k).

3 The distributor's seat: the hypothesis test

All prior DS-seat bounds were measured in the routing-fixed world; the thesis DRL lived in the as-shipped world, where the distributor's allocation drives the trust-routed HC's behavior — the only routing lever there. Testing the full decision-space grid there, on the thesis's own metric:

Reading: **simple distributor rules beat base stock decisively but reach only half of the thesis RL's claimed 89%**. The gap is attributable to (a) a mechanism outside the enumerated family — none identified — and/or (b) baseline non-comparability: Table 3.9 is unreplicated (our

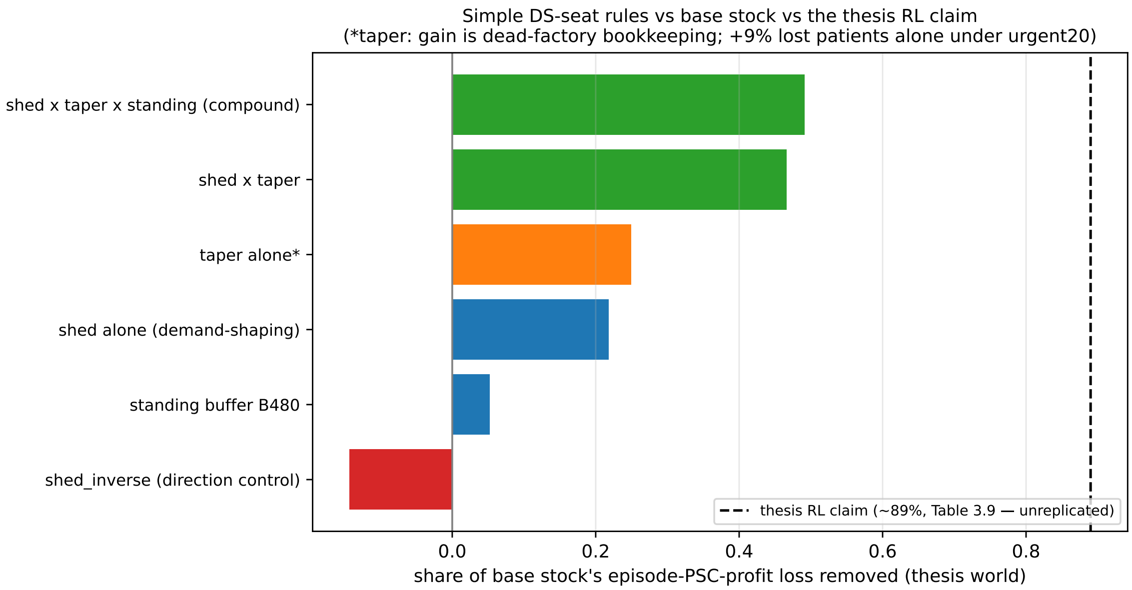

Figure 1: Share of base stock's episode-PSC-profit loss removed by simple distributor rules (thesis world, reporting seeds, audited). The compound reaches 49.1% (50.7% under urgent demand) with fill 0.858 vs. 0.808 and lost patients 1,575 vs. 1,768. The direction control (serve the trust-sensitive HC instead of the captive) is *worse than doing nothing* (−14.3%) — demand-shaping through the trust loop is real and directional. State-elevated ordering is absent because it is dead by supply physics: inventory cannot climb during the disruption (4 units measured); only the standing form pre-positions.

own corrected-pipeline DRL run *lost* to base stock), and its exact baseline configuration is the single most valuable piece of missing information (request pending with the thesis author). One caveat travels with the compound: the taper's contribution is dead-factory bookkeeping under the cost lens and costs +9% lost patients *when used alone* under urgent demand; the shed component offsets it.

4 Scarcity: the frontier, the robust knob, and two honest negatives

The robust knob is demand-aware buffer sizing (B = (measured demand − surviving deliverable) × duration): at sat50 under urgent demand it cuts lost patients to **41 vs. 505** at the frozen-grid optimum (dp-cost 1,726k vs. 1,865k, reporting seeds); its other endpoint (sat30 → B=4,800) independently matched the measured optimum.

Negative result, kept honest: the severity-aware redirect (cap rerouted volume at the surviving chain's headroom) fails in BOTH natural implementations — delivery-observation caps throttle the supplier's ramp (orders are the production signal in this simulator), and capacity-signal caps still under-order because over-ordering costs the orderer nothing here. The sat30 routing penalty is a delivery-mix problem, not an order-volume problem, and is fixed by buffer placement, not by order caps. "One robust routing policy across the map" does not exist in this class; per protocol the 9-cell sweep for the refuted policy was not run.

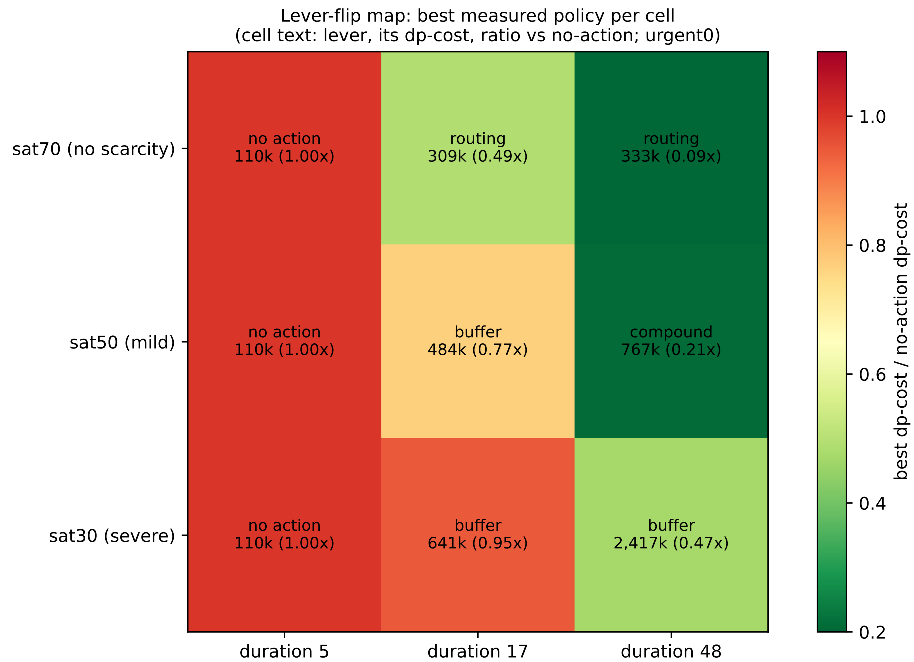

Figure 2: The lever-flip map (urgent0, reporting seeds), with the sat30/d48 cell CORRECTED by the location screen: at severe scarcity the buffer belongs AT THE DISRUPTED CHAIN (2,417k, −30% vs. the healthy location) — stock goes where the unservable demand is, once the surviving chain is too weak to absorb rerouting. The two boundaries stand: no response pays below the ∼6-period stock-cover duration; routing dominates with surviving headroom; the compound owns mild scarcity; inventory owns severe scarcity.

5 What we learned

1. The decomposition: **information sharing** (largest single item at thesis severity; two-line rule = oracle) + **routing** (dominant with surviving headroom) + **inventory, correctly LOCATED and demand-SIZED** (necessary under scarcity; location flips to the disrupted chain when the surviving chain is weak) + **true foresight** (∼5%, corrected from "nothing") − **over-reaction** (responses below stock-cover duration; sharp or capped rerouting under severe scarcity; clever reactive allocation outside the captive-shed case).

2. From the DS seat, **the user's hypothesis holds in spirit**: 3 parameters of simple rules beat base stock by half of what the RL claimed, with the residual unresolvable until the thesis baseline is known.

3. Cleverness keeps losing to structure: five reactive rules backfired; both redirect caps backfired; the two levers that worked everywhere are *where you put stock* and *what signal you share.*

4. Process: two contaminated batches were caught by gates/diagnostics and discarded; one rounding over-claim was caught by independent audit and corrected — the verification protocol earned its cost.

6 Next steps

1. **Obtain the thesis Table 3.9 configuration** (baseline included) — gates the interpretation of the 49%-vs-89% gap; everything else about the DS-seat story is settled.

2. **Recurring-disruption axis** (held, new regime): required pre-publication robustness check — premium amortization could move the map's buffer cells.

3. **Ramped onset** (held): the second paper — the one regime where detection (vs. info-sharing) could earn value, given the masking result.

4. **When the corrected DRL arrives**: regime-dependent bars, all simple — thesis world: the DS-seat compound (-49%); slack system: 273k; sat50: 767k–1,726k with demand-aware B. Framing: measure how close learning comes to the structured optimum.

5. Paper assembly: the three-lever decomposition with the lever-flip map and the DS-seat result as the RL-relevant centerpiece, all claims carrying their audit trail.